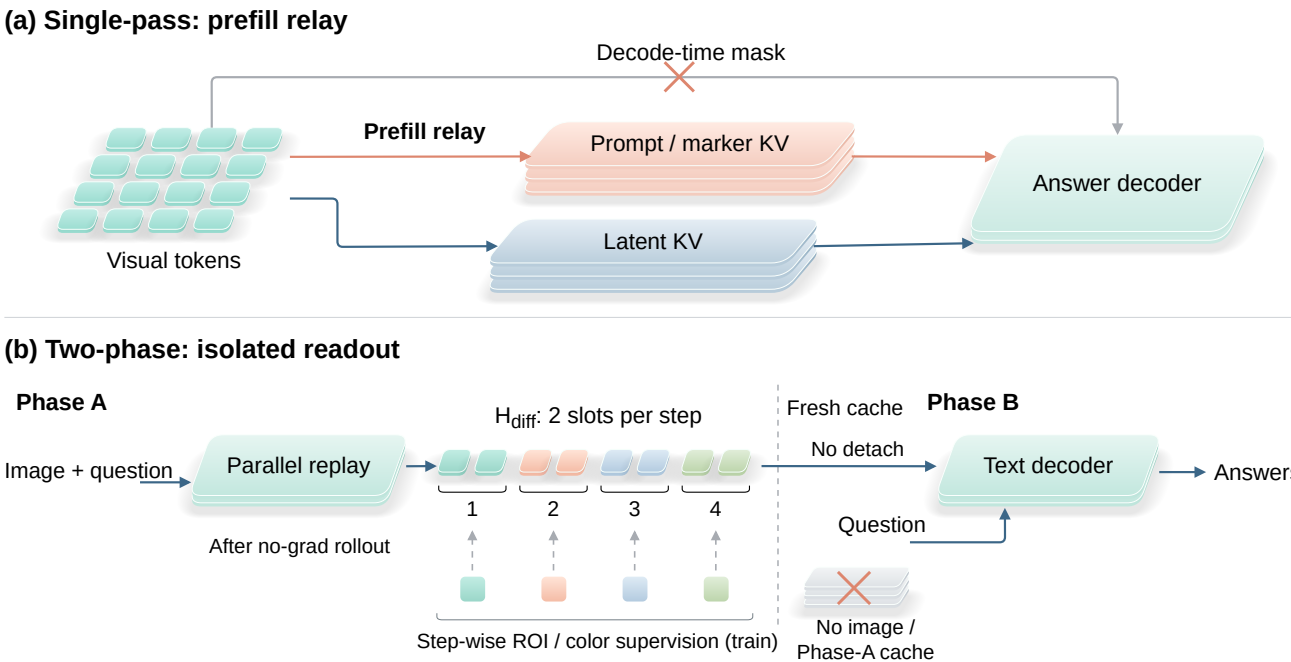

(a) Single-pass: prefill relay
Decode-time mask
Prefill relay
Prompt / marker KV
Answer decoder
Visual tokens
Latent KV
(b) Two-phase: isolated readout
Phase A
Image + question
Parallel replay
After no-grad rollout
H_diff: 2 slots per step
1
2
3
4
Step-wise ROI / color supervision (train)
Fresh cache
Phase B
No detach
Text decoder
Answers
Question
No image /
Phase-A cache